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Python 3 (ipykernel)

open_set = PriorityQueue()
open_set.put((0, start))
came_from = {}
g_score = {node: float('inf') for node in graph}
g_score[start] = 0

while not open_set.empty():
    _, current = open_set.get()

    if current == goal:
        path = []
        while current in came_from:
            path.append(current)
            current = came_from[current]
        path.append(start)
        return path[::-1]

    for neighbor, cost in graph[current]:
        tentative_g = g_score[current] + cost
        if tentative_g < g_score[neighbor]:
            came_from[neighbor] = current
            g_score[neighbor] = tentative_g
            f = tentative_g + heuristic[neighbor]
            open_set.put((f, neighbor))

    return None

# Define graph and heuristic
graph = {
    'A': [('B', 1), ('C', 4)],
    'B': [('D', 2), ('E', 5)],
    'C': [('F', 1)],
    'D': [('G', 2)],
    'E': [('G', 1)],
    'F': [('G', 3)],
    'G': []
}

heuristic = {
    'A': 7, 'B': 6, 'C': 2,
    'D': 1, 'E': 0, 'F': 3, 'G': 0
}

path = a_star(graph, 'A', 'G', heuristic)
print("Shortest path using A*:", path)

Shortest path using A*: ['A', 'B', 'D', 'G']
```